

Skill Arena - Phase 7 Document 4

Replay Intelligence, Match Integrity Analytics, Exploit Detection & Competitive Fairness Architecture

Document Purpose

Define how replay data is transformed into a competitive intelligence system for fairness, anti-cheat and platform protection.

Business Objective

Use replay information to continuously improve challenge integrity, detect exploits and protect platform sustainability.

Core Philosophy

Every match becomes a source of intelligence, not merely a dispute record.

Replay Intelligence Engine

Central platform responsible for analyzing historical and live replay data.

Replay Data Sources

PvP matches, House Challenges, Tournaments, Verification Challenges and Special Events.

Replay Metadata

Player IDs, Maze IDs, Completion Times, Route Paths, Decision Events and Validation Records.

Route Analysis Engine

Analyze how players move through challenges and compare behaviour against expected patterns.

Skill Measurement Framework

Measure decision quality, route efficiency, recovery ability and consistency.

Player Skill Profiles

Generate long-term player skill models independent of wins and losses.

Exploit Detection Engine

Detect challenge structures that produce abnormal success rates or unusual player behaviour.

Maze Leak Detection

Identify signs that challenge layouts may have been shared externally.

Route Similarity Analysis

Detect multiple players repeatedly following identical high-efficiency routes.

Replay Correlation Engine

Compare thousands of replays to discover coordinated behaviour patterns.

Bot Pattern Intelligence

Identify emerging automation strategies and behavioural signatures.

Macro Detection Support

Replay analysis contributes evidence to anti-bot and trust score systems.

Challenge Health Monitoring

Track challenge fairness, completion distributions and failure patterns.

Difficulty Validation

Confirm actual difficulty aligns with intended difficulty classifications.

Integrity Analytics

Measure fairness across categories, leagues, tournaments and house challenges.

House Protection Metrics

Identify challenges producing abnormal payouts or exploit-driven losses.

Player Protection Metrics

Identify challenges that may be excessively difficult or unfair.

Adaptive Feedback Loop

Replay intelligence feeds the AI Solver, Calibration Engine and House Risk Engine.

Investigation Support

Investigators receive replay evidence, route comparisons and behavioural analytics.

Tournament Integrity Monitoring

Additional replay analysis applied to championship and high-value events.

Historical Intelligence Repository

Store replay intelligence for trend analysis and long-term investigations.

Monitoring Requirements

Continuous monitoring of route behaviour, challenge performance and exploit indicators.

Audit Requirements

Replay analyses, intelligence findings and enforcement actions permanently logged.

Future Expansion

Supports future games, AI coaching, advanced analytics and automated exploit discovery.

Acceptance Criteria

Replay systems actively improve platform fairness, challenge quality and operational intelligence.